

Part 1: <https://studio.edgeimpulse.com/public/1074052/live>

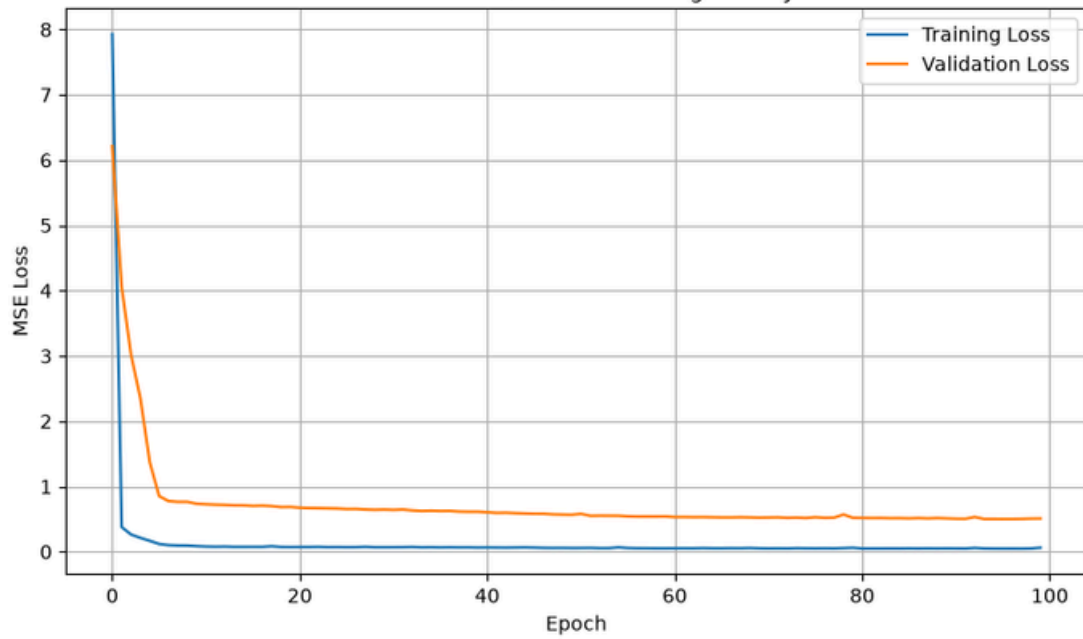
1. Raw inputs: accX, accY, accZ.
2. NNClassifier features:
 - a. Type: Spectral features
 - b. 33 input features
 - c. 5 features: accX RMS, accX Peak 1 Height, accY Peak 3 Height, accX Peak 1 Freq, accY RMS
3. NNClassifier outputs: nominal, off
4. Anomaly detector inputs:
 - a. Type: selected subset of NNClassifier outputs:
 - b. 4 input features
 - c. accX RMS, accX Peak 1 Height, accY Spectral power 2.0-5.0 Hz, accZ Spectral power 2.0-5.0 Hz
5. Deployment evidence:

```
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 593 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.683594
  off: 0.316406
Anomaly prediction: 96.062843
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 593 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.984375
  off: 0.015625
Anomaly prediction: 404.828522
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 594 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.554687
  off: 0.445312
Anomaly prediction: 381.407166
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 610 us, anomaly 524 us, postprocessing 5 ms
#Classification predictions:
  nominal: 0.500000
  off: 0.500000
Anomaly prediction: 16.565990
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 609 us, anomaly 528 us, postprocessing 5 ms
#Classification predictions:
  nominal: 0.500000
  off: 0.500000
Anomaly prediction: 16.564438
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 608 us, anomaly 526 us, postprocessing 990 us
#Classification predictions:
  nominal: 0.500000
  off: 0.500000
Anomaly prediction: 16.562874
Starting inferencing in 2 seconds...
```

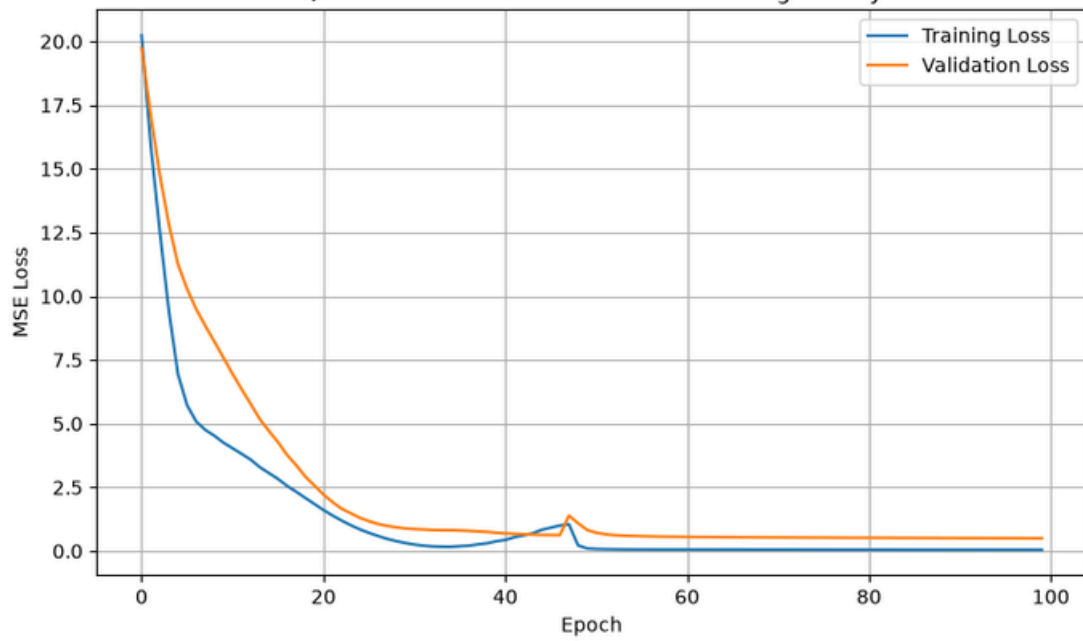
6. The classifier should not always be trusted. There were times when it had a 50/50 split and did not choose between either class. There were also times when it was very confident that it was nominal. I tested the deployment by shaking the board like a fan would. The anomaly score also changed drastically, ranging from 16 to 400. The classifier can produce misleading results if it is being given input of something other than a nominal or off fan.

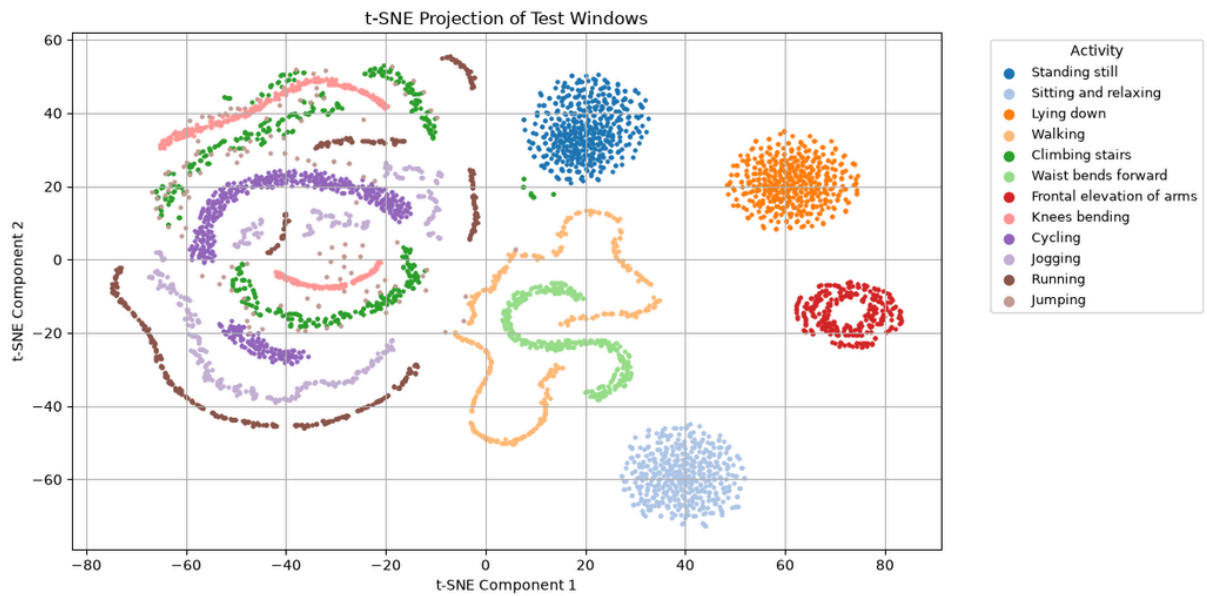
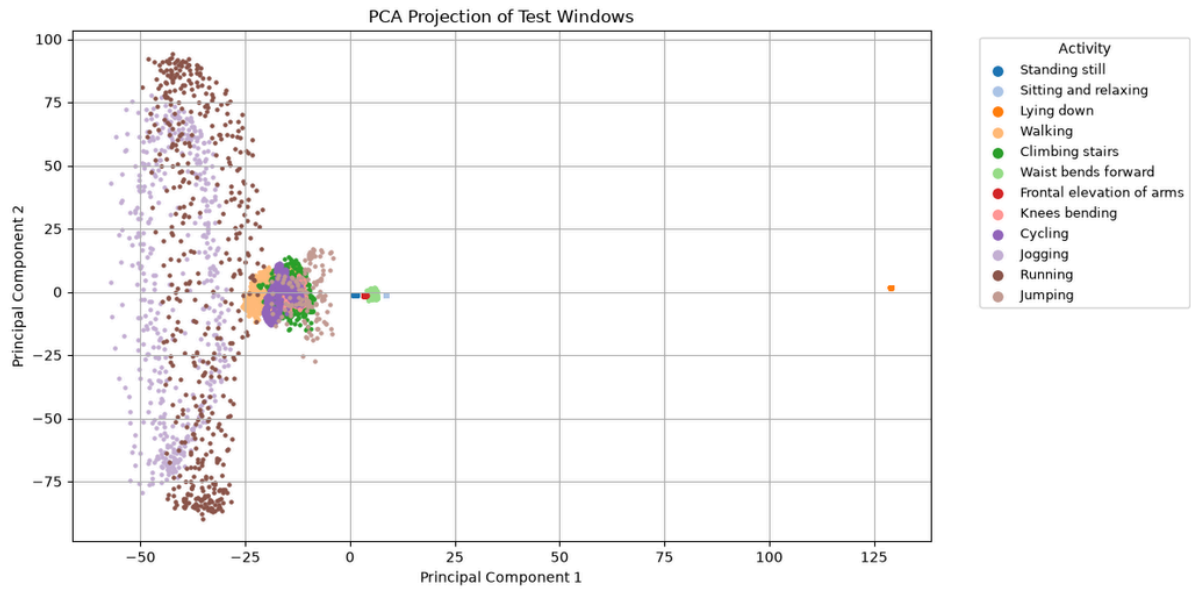
Part 2:

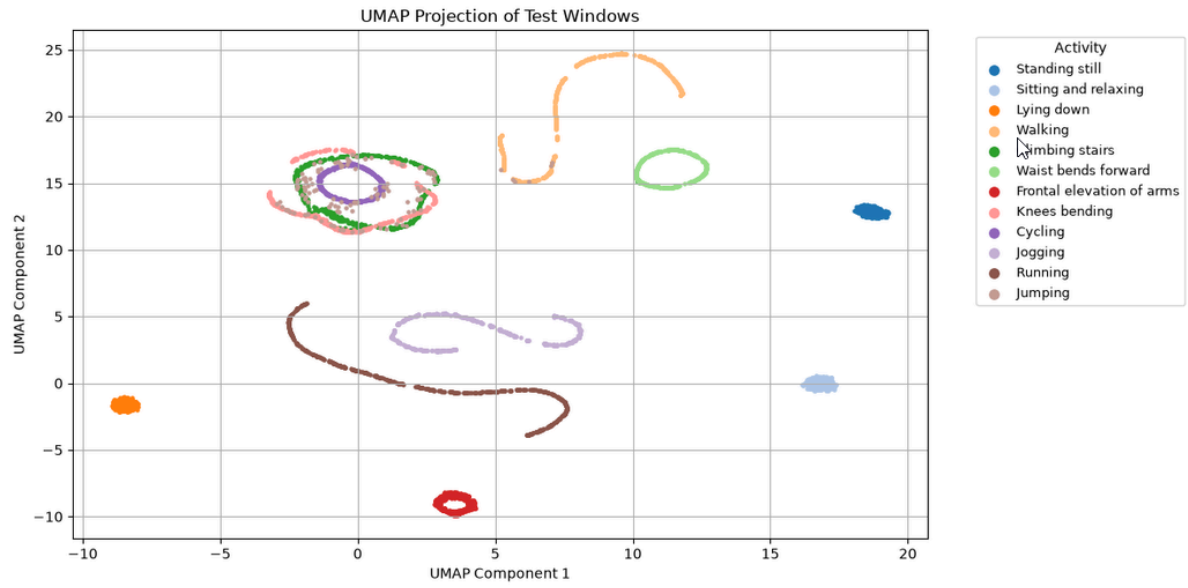
Float32 Autoencoder Training History

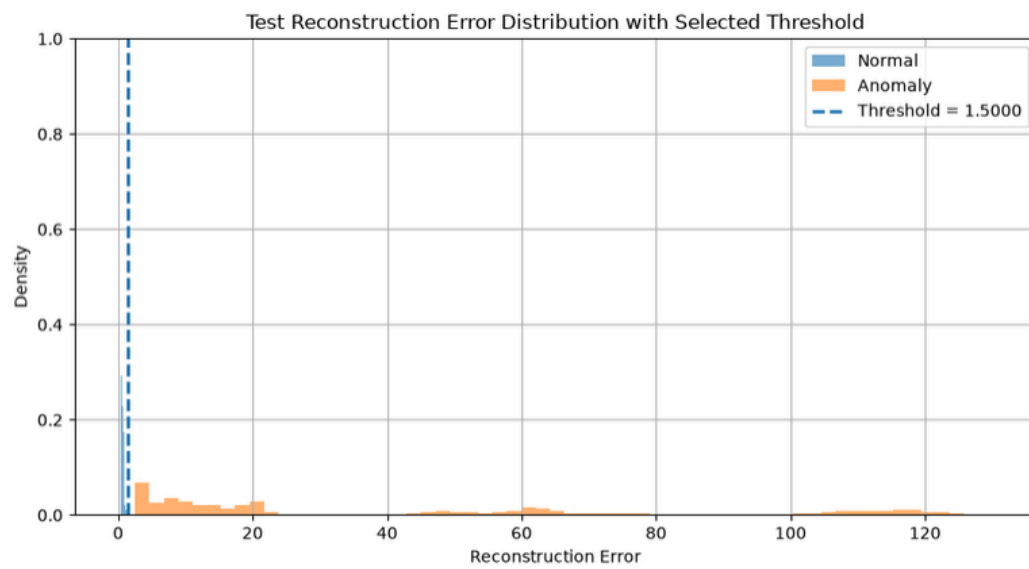


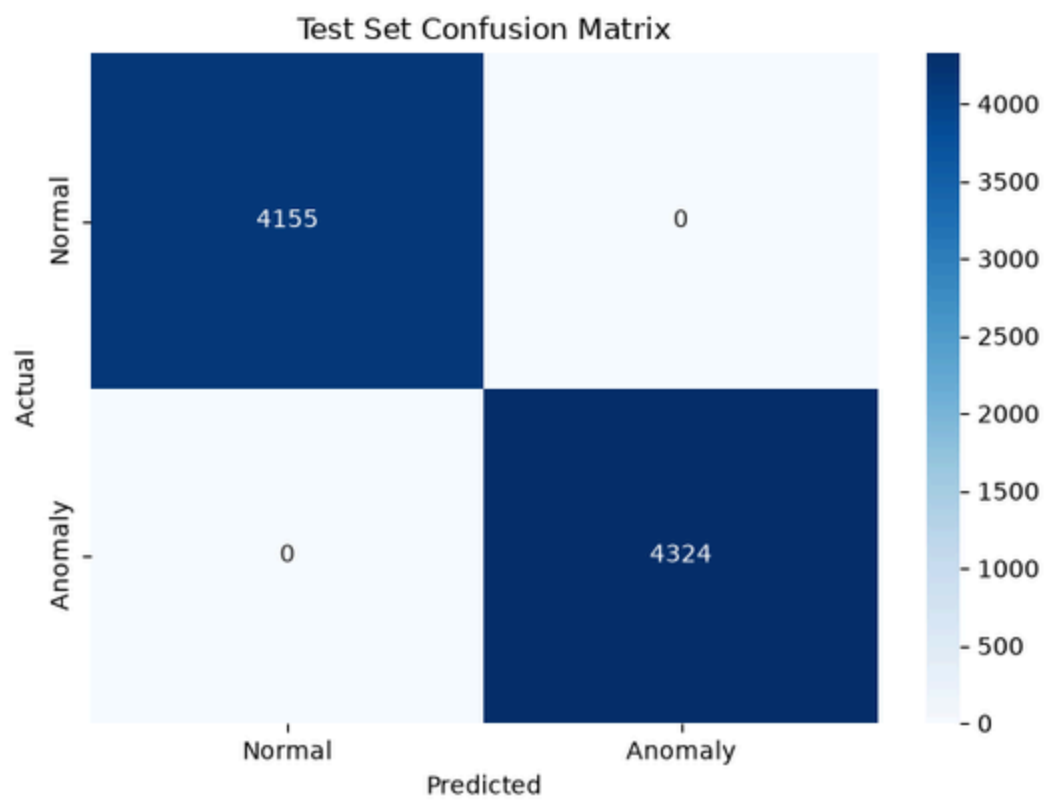
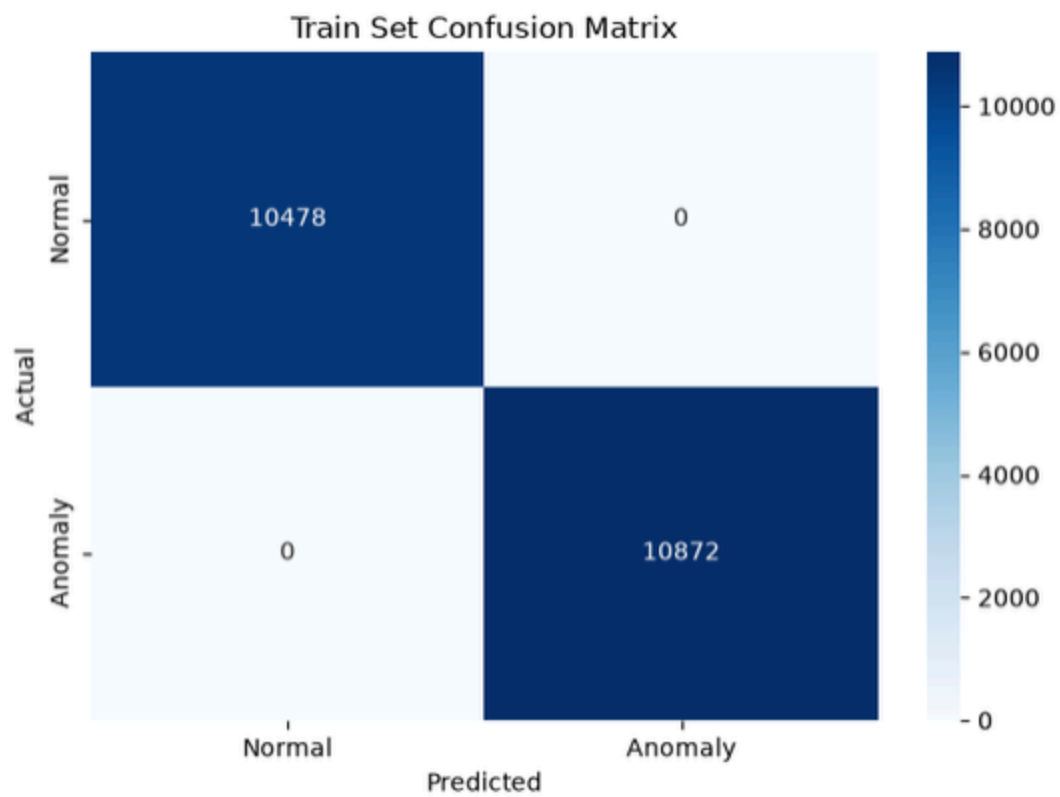
Quantization-Aware Autoencoder Training History











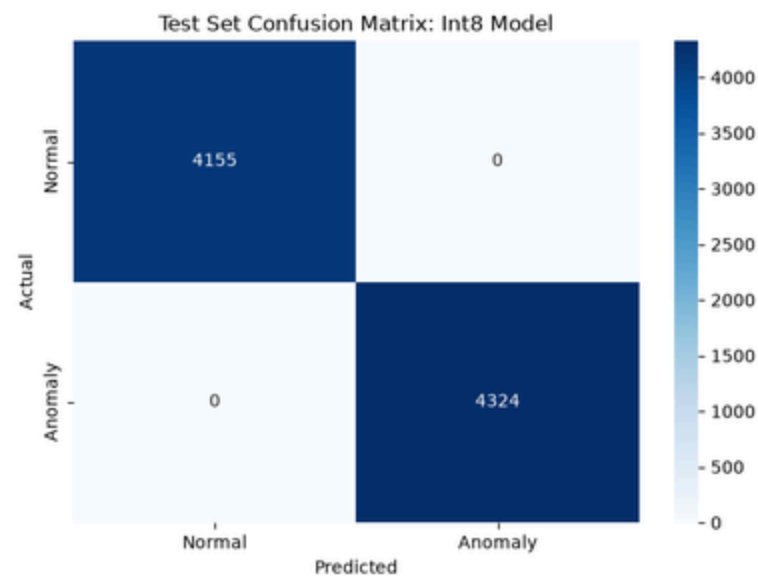
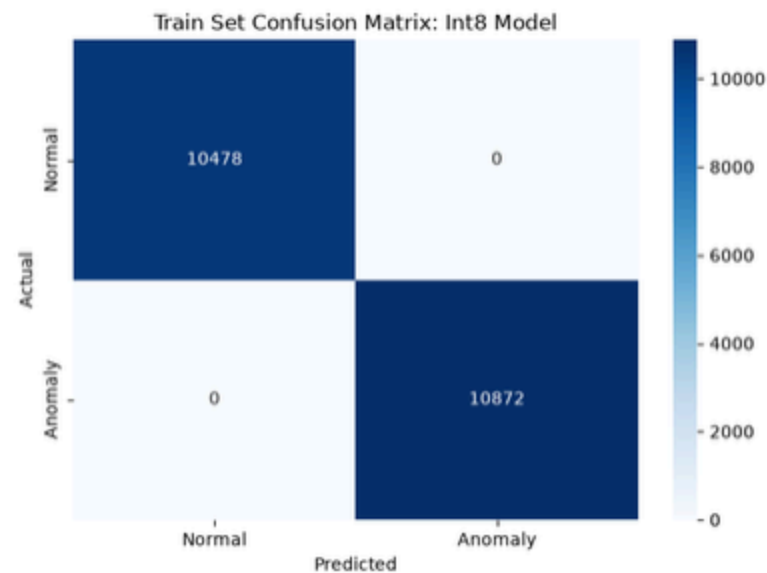
INFO: Created tensorflow lite ANNPALK delegate for CPU.

Train set evaluation: int8 model

	precision	recall	f1-score	support
Normal (0)	1.00	1.00	1.00	10478
Anomaly (1)	1.00	1.00	1.00	10872
accuracy			1.00	21350
macro avg	1.00	1.00	1.00	21350
weighted avg	1.00	1.00	1.00	21350

Test set evaluation: int8 model

	precision	recall	f1-score	support
Normal (0)	1.00	1.00	1.00	4155
Anomaly (1)	1.00	1.00	1.00	4324
accuracy			1.00	8479
macro avg	1.00	1.00	1.00	8479
weighted avg	1.00	1.00	1.00	8479



```
Output  Serial Monitor X
Message (Enter to send message to 'Arduino Nano 33 BLE' on 'COM4')
Reconstruction error: 0.027735
Result: Normal activity
Reconstruction error: 0.027723
Result: Normal activity
Reconstruction error: 0.027864
Result: Normal activity
Reconstruction error: 0.027994
Result: Normal activity
Reconstruction error: 0.027936
Result: Normal activity
Reconstruction error: 0.027807
Result: Normal activity
Reconstruction error: 0.027740
Result: Normal activity
Reconstruction error: 0.027744
Result: Normal activity
Reconstruction error: 0.027675
Result: Normal activity
Reconstruction error: 0.027633
Result: Normal activity
Reconstruction error: 0.027604
Result: Normal activity
Reconstruction error: 0.027564
Result: Normal activity
Reconstruction error: 0.027534
Result: Normal activity
Reconstruction error: 0.027482
Result: Normal activity
Reconstruction error: 0.027427
Result: Normal activity
Reconstruction error: 0.027400
Result: Normal activity
```

Part 3:

1. I chose 1.5 as the threshold. The histogram of reconstruction errors had the normal-window errors close to zero and anomaly errors starting around four. I chose 1.5 because it was right in the middle of the gap.
2. The autoencoder is trained only on normal activity windows, so it learns to compress and reconstruct the patterns of normal movement well. When we give it a window of anomalous movement, it does not know how to reconstruct it, so it has a large error. Low error means the input looks like something the model knows, while high error means it does not fit the learned pattern.
3. The window size and stride must be preserved. The feature columns and sensor placement, and how they were normalized, and the chosen threshold must all be the same. The Arduino has to reproduce the same input pipeline that the model was trained on.
4. The model learned patterns for inputs that were formatted and scaled the way that the training data was. If the Arduino sketch windowed the sensor data differently or changed the order of the axes, the input that the model sees during inference won't match what it learned, and will produce meaningless reconstruction errors.

5. The main limits are memory, so only small models fit. This can degrade the accuracy and change the error distribution, which would require a new threshold. A single threshold might also not generalize across different uses.

```
Output  Serial Monitor X
Message (Enter to send message to 'Arduino Nano 33 BLE' on 'COM4')
Reconstruction error: 2.512733
Result: Anomaly detected
Reconstruction error: 2.513144
Result: Anomaly detected
Reconstruction error: 2.518346
Result: Anomaly detected
Reconstruction error: 2.589924
Result: Anomaly detected
Reconstruction error: 2.605741
Result: Anomaly detected
Reconstruction error: 2.720932
Result: Anomaly detected
Reconstruction error: 2.873949
Result: Anomaly detected
Reconstruction error: 2.877631
Result: Anomaly detected
Reconstruction error: 2.923943
Result: Anomaly detected
Reconstruction error: 3.005363
Result: Anomaly detected
Reconstruction error: 3.011018
Result: Anomaly detected
Reconstruction error: 3.001587
Result: Anomaly detected
Reconstruction error: 3.021361
Result: Anomaly detected
Reconstruction error: 3.042822
Result: Anomaly detected
Reconstruction error: 3.105474
Result: Anomaly detected
Reconstruction error: 3.116227
Result: Anomaly detected
```